

Note 5.6 The Gram-Schmidt Orthogonalization Process

Intro

The process is for constructing an orthonormal basis for an n -dimensional inner product space V . It uses projections to transform an ordinary basis into an orthonormal basis.

Theorem 5.6.1 The Gram-Schmidt Process

Let $\{\mathbf{x}_1, \dots, \mathbf{x}_n\}$ be a basis for V , then let

$$\mathbf{u}_1 = \left(\frac{1}{\|\mathbf{x}_1\|} \right) \mathbf{x}_1$$

and define the rest vectors recursively by

$$\mathbf{u}_{k+1} = \frac{1}{\|\mathbf{x}_{k+1} - \mathbf{p}_k\|} (\mathbf{x}_{k+1} - \mathbf{p}_k)$$

where

$$\mathbf{p}_k = \langle \mathbf{x}_{k+1}, \mathbf{u}_1 \rangle \mathbf{u}_1 + \dots + \langle \mathbf{x}_{k+1}, \mathbf{u}_k \rangle \mathbf{u}_k$$

is the projection of $\mathbf{x}_{k+1}$ onto $\text{Span}(\mathbf{u}_1, \dots, \mathbf{u}_n)$ and the set

$$\{\mathbf{u}_1, \dots, \mathbf{u}_n\}$$

is an orthonormal basis for V .

Theorem 5.6.2 Gram-Schmidt QR Factorization

For an $m \times n$ matrix A with rank n , it can be factorized into a product QR , where Q is an $m \times n$ matrix with orthonormal column vectors and R is an upper triangular $n \times n$ matrix whose diagonal entries are all positive.

For Q , its column vectors are $\mathbf{u}_1, \dots, \mathbf{u}_n$. For R , its elements are

$$r_{11} = \|\mathbf{a}_{11}\|, r_{kk} = \|\mathbf{a}_k - \mathbf{p}_{k-1}\| (k = 2, \dots, n)$$

$$r_{ij} = \langle \mathbf{q}_i, \mathbf{a}_j \rangle (i = 1, \dots, j-1; j = 2, \dots, n)$$

That is, elements on the ii diagonal are the norms of vectors perpendicular to $\text{Span}(\mathbf{u}_1, \dots, \mathbf{u}_i)$. Elements on j th column are the coordinates of $\mathbf{p}_j$ with respect to $\text{Span}(\mathbf{u}_1, \dots, \mathbf{u}_j)$.

Theorem 5.6.3

For an $m \times n$ matrix A with rank n , the least square solution of $A\mathbf{x} = \mathbf{b}$ is

$$\hat{\mathbf{x}} = R^{-1}Q^T\mathbf{b}$$

The Modified Gram-Schmidt Process

For the first element,

$$\mathbf{q}_1 = \frac{1}{\|\mathbf{a}_1\|} \mathbf{a}_1$$

Then after computing k th vector, we need to modify the latter vectors by minus the original vector's projection onto $\mathbf{q}_k$ such that the latter vectors are orthogonal to previous orthonormal set. And the k th element can be obtained by

$$\mathbf{q}_k = \frac{1}{\|\mathbf{a}_k^{(k-1)}\|} \mathbf{a}_k^{(k-1)}$$

where the $\mathbf{a}_k^{(k-1)}$ is the vector after modifying $k - 1$ times.